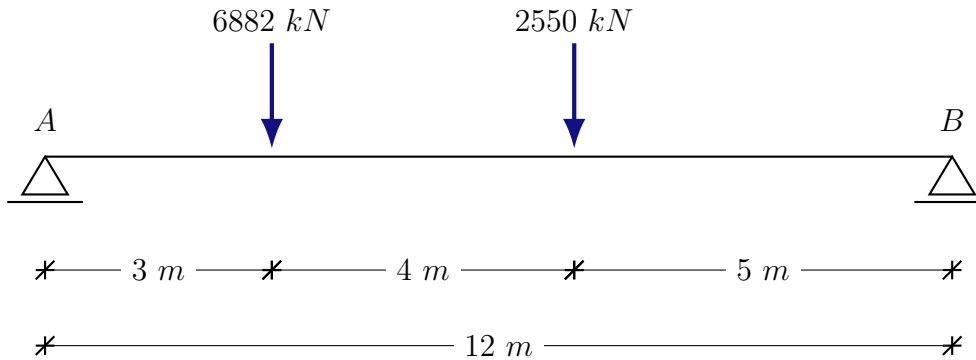


Exercício

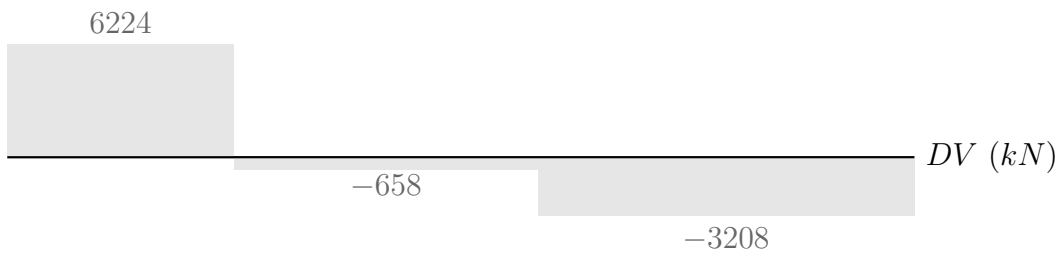
1. Dimensione a viga AB usando aço AR 350 COR, as cargas já estão majoradas escolha um perfil VS.



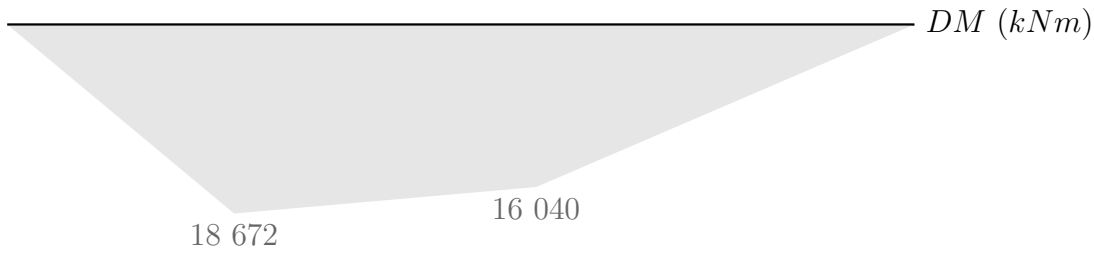
Reações de apoio e esforços solicitantes

$$\begin{aligned}\sum M_B &= 0 \\ -12V_A + 9 \cdot 6882 + 5 \cdot 2550 &= 0 \\ V_A &= 6224 \text{ kN}\end{aligned}$$

$$\begin{aligned}\sum F_y &= 0 \\ V_A + V_B - 6882 - 2550 &= 0 \\ V_B &= 3208 \text{ kN}\end{aligned}$$



$$V_{Sd,max} = V_A = 6224 \text{ kN}$$



$$M_{Sd,max} = M_C^1 = 18\,672 \text{ kNm}$$

Propriedades do material e coeficiente

AR 350 COR $\rightarrow f_y = 350 \text{ MPa}, f_u = 485 \text{ MPa}$ (Tabela A.1 da NBR 8800)

$\gamma_{a1} = 1,1$ (Tabela 3 da NBR 8800)

Pré-dimensionamento à flexão

$\lambda \leq \lambda_p$ (Anexo D.2.1 da NBR 8800)

$M_{Sd} \leq M_{Rd}$ (Item 5.4.1.3 da NBR 8800)

$M_{Sd} \leq \frac{1}{\gamma_{a1}} M_{pl} \rightarrow M_{pl} = f_y W$ (Tabela D.1 da NBR 8800)

$$W = \frac{M_{Sd} \cdot \gamma_{a1}}{f_y}$$

$$W = \frac{18\,672 \cdot 10^3 \times 1,1}{350 \cdot 10^6}$$

$$W \approx 58,68 \cdot 10^{-3} \text{ m}^3$$

Pré-dimensionamento ao cisalhamento

$\lambda \leq \lambda_p$ (Anexo D.2.1 da NBR 8800)

$V_{Sd} \leq V_{Rd}$ (Item 5.4.1.3 da NBR 8800)

$V_{Sd} \leq \frac{V_{pl}}{\gamma_{a1}} \rightarrow V_{pl} = 0,6 A_w f_y$ (Item 5.4.3.1.1, 5.4.3.1.2 da NBR 8800)

$$A_w = \frac{V_{Sd} \times \gamma_{a1}}{0,6 f_y}$$

$$A_w = \frac{6224 \cdot 10^3 \times 1,1}{0,6 \times 350 \cdot 10^6}$$

$$A_w = 32,60 \cdot 10^{-3} \text{ m}^2$$

Perfil VS

$A_w = d \times t_w$ (Item 5.4.3.1.2 da NBR 8800)

$$A_w = 1500 \cdot 10^{-3} \times 12,5 \cdot 10^{-3}$$

$$A_w = 18,75 \cdot 10^{-3} \text{ m}^2$$

$$VS\ 1500 \times 492 \rightarrow W = 38950 \text{ cm}^3, A = 18,75 \cdot 10^{-3} \text{ m}^2$$

∴ Não existe perfil VS capaz de suportar esse carregamento. Seria necessário projetar uma sob medida.

CORREÇÃO

Pré dimensionamento à flexão

$$\begin{aligned} M_{Sd} &\leq M_{Rd} \\ 18672 \cdot 10^3 &\leq \frac{W \times 350 \cdot 10^6}{1,1} \\ W &\approx 58,68 \cdot 10^{-3} \text{ m}^3 \end{aligned}$$

Perfil escolhido: VS $800 \times 143 \rightarrow Z = 5910 \text{ cm}^3$

FLM

$$\begin{aligned} \lambda &\leq \lambda_p \\ \frac{b}{t} &\leq 0,38 \sqrt{\frac{E}{f_y}} \\ \frac{\frac{320 \cdot 10^{-3}}{2}}{19 \cdot 10^{-3}} &\leq 0,38 \sqrt{\frac{200\,000 \cdot 10^6}{350 \cdot 10^6}} \\ 8,42 &\leq 9,08 \end{aligned}$$

$$M_{Rd} = \frac{M_{pl}}{\gamma_{a1}} = \frac{5910 \times 35}{1,1} = 188045,45 \text{ kNcm}$$

FLA

$$\lambda = \frac{h}{t_w} = \frac{76,2}{0,8} = 95,25$$

$$\lambda_p = 3,76 \sqrt{\frac{20000}{35}} = 89,88$$

$$\lambda_r = 5,7 \sqrt{\frac{20000}{35}} = 136,25$$

$$\lambda_p \leq \lambda < \lambda_r$$

$$M_r = w_x \times f_y = 5374 \times 35 = 188090 \text{ kNcm}$$

$$M_{pl} = 206850 \text{ kNcm}$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} = (M_{pl} - (M_{pl} - M_r) \frac{\lambda - \lambda_p}{\lambda_r - \lambda_p}) = 186070,53 \text{ kNcm}$$

$\therefore$ Acontecerá flambagem na alma desse perfil.

SEGUNDA TENTATIVA

FLM

$0,35 \leq k_c \leq 0,76$ (Tabela 4, nota de rodapé a da Norma 8800)

$$k_c = \frac{4}{\sqrt{\frac{h}{t_w}}}$$
$$k_c = \frac{4}{\sqrt{\frac{81,2}{0,8}}}$$
$$k_c = 0,397$$

$$\lambda_p < \lambda \leq \lambda_r$$

$$0,38 \sqrt{\frac{E}{f_y}} < \frac{b}{t} \leq 0,95 \sqrt{\frac{E}{(f_y - \sigma_r)/k_c}}$$
$$0,38 \sqrt{\frac{200\,000 \cdot 10^6}{350 \cdot 10^6}} < \frac{\frac{350 \cdot 10^{-3}}{2}}{19 \cdot 10^{-3}} \leq 0,95 \sqrt{\frac{200\,000 \cdot 10^6}{(350 \cdot 10^6 - \sigma_r)/0,397}}$$
$$9,08 < 9,21 \leq 17,10$$

$$M_{pl} = 6845 \times 35 = 2396 \text{ kNcm}$$

$$M_r = 6243 \times 35 = 218505 \text{ kNcm}$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} = (M_{pl} - (M_{pl} - M_r) \frac{9,21 - 9,08}{17,12 - 9,08}) = 217485 \text{ kNcm}$$

FLA

$$\lambda = \frac{b}{t_w} = \frac{81,2}{0,8} = 101,5$$

$$\lambda_p = 89,88$$

$$\lambda_r = 136,25$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} = (M_{pl} - (M_{pl} - M_r) \frac{9,21 - 9,08}{17,12 - 9,08}) = 212995 \text{ kNcm}$$

FLT

$$\lambda = \frac{L_b}{r_y} = \frac{1200 \cdot 10^{-2}}{8,28 \cdot 10^{-2}} = 144,92$$

$$\lambda = 1,76 \sqrt{\frac{E}{f_y}} = 42,07$$

$$\lambda_r = \frac{1,38 c_b \sqrt{I_y J}}{r_y \times J \times B_y} \sqrt{1 + \sqrt{\frac{1 + 27 C_w B_1^2}{C_b^2}}} = 118$$

$$B_1 = \frac{(f_y - \sigma_r)W}{EJ}$$

$$B_1 = \frac{(\frac{70}{100}f_y)6243}{20\,000 \times 174}$$

$$B_1 = 0,0439$$

$$C_w = \frac{I_y(d - t_y)^2}{4} \text{ Para seções I}$$

$$C_w = \frac{13581(85 - 1,9)^2}{4}$$

$$C_w = 2344672,35$$

$$C_b d = \frac{12,5M_{max}}{2,5M_{max} + 3M_A + 4M_B + 3M_C} R_m \rightarrow R_M = 1$$

$$C_b = \frac{12,5 \times 186720}{2,5 \times 186720 + 3 \times 186720 + 4 \times 16698 + 3 \times 9624} R_m$$

$$C_b = 1,176$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} M_{Ce}$$

text